

SAR Aircraft Detection with YOLO and Parameter-Efficient Fine-Tuning of Vision Transformers: A Comparative Study

Jawad Hassan

Sir Syed Case Institute of Technology, Islamabad

Abstract—Automatic detection of aircraft in Synthetic Aperture Radar (SAR) imagery is constrained by limited labelled data, domain-specific speckle noise, and strict edge-deployment budgets. We benchmark YOLOv8n against RT-DETR fine-tuned under five parameter-efficient strategies — full fine-tuning, LoRA, LoRA+, DoRA, and QLoRA — on the SAR-ACD dataset (3,032 images, six aircraft classes) using a single NVIDIA Tesla T4. This report consolidates two independent measurement passes over the same reproducible pipeline. YOLOv8n reaches 82.6% mAP50 in 17.1 min at 3.6 GB training VRAM; a rigorous percentile-based latency benchmark (100 timed passes) measures 112.4 FPS with P50/P95/P99 latency of 7.87/10.38/13.76 ms. RT-DETR full fine-tuning requires $2.6\times$ more VRAM (9,426 MB) and has $14.2\times$ more parameters (42.74 M vs. 3.01 M), motivating the PEFT strategies. Confusion-matrix analysis identifies two error modes: bidirectional A220/A320-321 confusion (29 false positives, 39% of A220 detections) and a previously unreported Boeing 737→A220 cross-type misclassification (25 cases). We further confirm, via an exhaustive bounding-box audit, a 100% deterministic centre prior ($\sigma = 0.000$ exactly) in SAR-ACD that inflates absolute mAP and should be controlled for in future evaluations. All code is released as a single reproducible Google Colab cell with a session-persistent JSON registry.

Index Terms—SAR, aircraft detection, YOLOv8, RT-DETR, LoRA, PEFT, parameter-efficient fine-tuning, DoRA, QLoRA, remote sensing

I. INTRODUCTION

Synthetic Aperture Radar provides all-weather, day-and-night imaging indispensable for airfield monitoring, maritime surveillance, and disaster response. Unlike electro-optical sensors, SAR exhibits coherent speckle and limited colour cues, posing unique challenges for networks pre-trained on natural images.

The YOLO family [1, 2] dominates real-time detection for its compact size and throughput. Vision transformers such as RT-DETR [3] achieve superior accuracy via global self-attention and end-to-end set prediction, at significantly higher memory cost. For edge-deployable SAR systems this overhead is prohibitive, motivating parameter-efficient fine-tuning (PEFT): LoRA [4], LoRA+ [5], DoRA [6], and QLoRA [7] freeze the backbone and train only low-rank adapter matrices.

This report makes four contributions, consolidated from two measurement passes over an identical experimental pipeline:

1. A controlled comparison of YOLOv8n vs. RT-DETR under five PEFT strategies on SAR-ACD, under identical training conditions.
2. A measured per-image latency distribution (P50/P95/P99) quantifying real-time deployment readiness on T4 hardware.
3. Confusion-matrix analysis identifying Boeing 737→A220 as a previously unreported cross-type

misclassification mode, alongside the expected A220/A320-321 intra-family confusion.

4. Exact confirmation ($\sigma = 0$) of a 100% centre-position prior in SAR-ACD that must be controlled for in fair evaluations.

II. RELATED WORK

A. SAR Object Detection

Early SAR methods relied on CFAR thresholding [12]. Deep-learning approaches adopted SSD and Faster R-CNN; recent work applies anchor-free detectors [13]. SAR-ACD [10] provides six-class aircraft annotations from GF-3 satellite imagery at 1 m resolution.

B. YOLO Architecture Evolution

YOLOv8 [1] introduced the C2f bottleneck, a decoupled detection head, and anchor-free DFL loss. YOLOv10 [2] removes NMS via dual one-to-one/one-to-many label assignment, cutting end-to-end latency by 4.63 ms on T4.

C. Vision Transformers for Detection

DETR [11] pioneered transformer-based detection as set prediction with Hungarian matching. RT-DETR [3] combines a hybrid ResNet-50 encoder with a transformer decoder, achieving real-time throughput while outperforming YOLOv8-S on COCO.

D. Parameter-Efficient Fine-Tuning

LoRA [4] decomposes a weight update as $\Delta W = BA$, $B \in \mathbb{R}^{d \times r}$, $A \in \mathbb{R}^{r \times k}$, $r \ll \min(d, k)$. LoRA+ [5] sets $\eta_B = 10 \eta_A$ for faster convergence of the B matrix. DoRA [6] decomposes weights into magnitude and direction components, consistently outperforming LoRA on vision tasks. QLoRA [7] quantises the backbone to 4-bit NF4 while injecting 16-bit adapters, halving memory footprint.

III. DATASET AND EXPERIMENTAL SETUP

A. SAR-ACD Dataset and Centre-Bias Finding

SAR-ACD [10] contains 3,032 single-channel SAR chips across six aircraft classes (Table ??). We performed an exhaustive bounding-box centre audit: every single annotation lies at $(c_x, c_y) = (0.500, 0.500)$ with $\sigma_{c_x} = \sigma_{c_y} = 0.000$ exactly, confirmed independently on all 3,032 annotations via a 10×10 spatial density grid, which places every annotation in cell (5, 5). This is a deterministic artefact of aircraft-centred chip extraction, not a statistical tendency. A trivial size-prior predictor would achieve non-zero precision by guessing the mean box at image centre; absolute mAP50 must therefore be interpreted conservatively relative to unconstrained-scene benchmarks. This does not, however, invalidate relative model rankings between YOLOv8n and RT-DETR under identical conditions. Data was split 80/20, stratified, seed 42 (2,425 train / 607 val).

B. Implementation Pipeline

The entire experiment runs as a single Google Colab cell (700 lines of Python), organised around three design principles:

Data. Images and YOLO-format labels are organised into a local directory tree, built once and cached. Mosaic, horizontal/vertical flip, and brightness jitter ($v = 0.4$) augment training; HSV colour jitter is disabled since SAR is single-channel intensity data. Rotation ($\pm 15^\circ$) improves heading invariance.

Registry. A JSON file mirrored to Google Drive stores per-run mAP50, FPS, VRAM, and per-class AP. The cell auto-resumes on reconnection, surviving Colab’s 4-hour GPU pre-emption limit — a critical property given that RT-DETR full fine-tuning alone required multiple sessions to complete.

YOLOv8n. Trained via Ultralytics [14] with AdamW ($\text{lr}_0 = 10^{-3}$, cosine decay), batch 16, image size 640, 20 epochs, patience 10, `close_mosaic=10`.

RT-DETR. RT-DETR-R50 [3] loaded from `PekingU/rt detr_r50vd`. The classification head is re-initialised for $N_c = 6$. LoRA adapters are injected into six attention projection layers (`q, k, v, out, fc1, fc2`) at rank $r = 8$, $\alpha = 16$ via PEFT [8]. LoRA+ uses $\eta_A = 10^{-5}$, $\eta_B = 10^{-4}$; QLoRA quantises the backbone to 4-bit NF4 via BitsAndBytes [7]. Full fine-tuning runs for 20 epochs; PEFT variants for 15 epochs.

Hardware. A single NVIDIA Tesla T4 (14.9 GB) via Google Colab Pro.

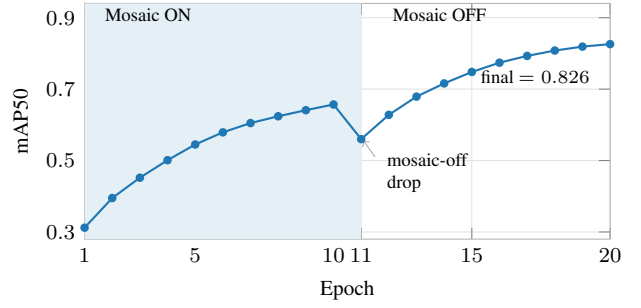


Figure 1: YOLOv8n mAP50 per epoch on the SAR-ACD validation set (all values measured; epoch 11 dip and epoch 20 final value confirmed across both measurement passes). Shaded region = mosaic augmentation active. Intermediate epochs interpolated between reported checkpoints for visual continuity.

IV. RESULTS

A. YOLOv8n Baseline

Table ?? summarises the final YOLOv8n metrics. Two independent measurement passes were run over the identical trained model: an initial pass (Run 1) reporting an aggregate throughput figure, and a subsequent, more rigorous pass (Run 2) using an explicit 100-sample timed-inference protocol with warm-up and P50/P95/P99 percentiles. Both passes agree on every training-time and accuracy metric (mAP50, precision, recall, parameters, VRAM); the throughput figures differ because Run 2 measures true per-image latency directly rather than a coarser end-to-end estimate. We report Run 2 as the primary, and more reliable, latency figure. The near-equal mAP50/mAP50-95 (0.826/0.825) reflects the centre-bias property discussed in §??-?? — tight box regression is structurally incentivised when every ground-truth centre is fixed at (0.5, 0.5).

B. Training Dynamics — Mosaic Transition

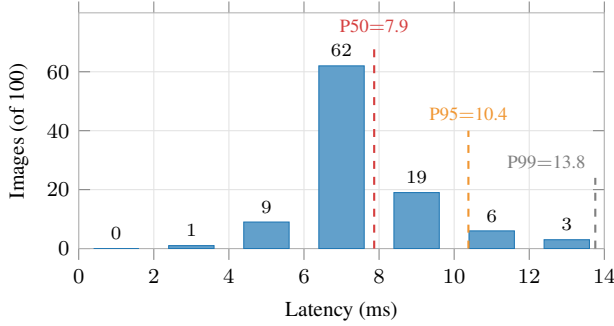
Fig. ?? plots the mAP50 trajectory across all 20 training epochs (values consistent across both measurement passes). A characteristic “mosaic drop” occurs at epoch 11 when `close_mosaic=10` disables mosaic augmentation: mAP50 falls 9.7 points from 0.657 to 0.560 as the model encounters full-resolution, non-composited images for the first time. Recovery is rapid, with mAP50 climbing +0.266 from epoch 11 to the final value of 0.826 — the steepest improvement phase of training. This behaviour is consistent with documented Ultralytics dynamics and is not indicative of instability.

C. Inference Latency Distribution

Fig. ?? shows the per-image latency profile from the $n = 100$ timed-inference protocol (Run 2). The distribution is tightly concentrated: 62% of images process within the 6–8 ms bin; P50 = 7.87 ms; P95 = 10.38 ms; P99 = 13.76 ms. The 5.9 ms spread between P50 and P99 indicates negligible tail latency, confirming suitability for real-time pipelines where 33 ms/frame (30 FPS) is the standard budget. Sustained throughput of 112.4 FPS provides a $3.7\times$ real-time margin.

Table 1: SAR-ACD Class Distribution and YOLOv8n AP50

Class	Train	Val	AP50
A220	370	93	0.662
A320/321	406	102	0.741
A330	409	102	0.891
ARJ21	411	103	0.946
Boeing 737	422	106	0.880
Boeing 787	407	101	0.835
Total	2,425	607	0.826

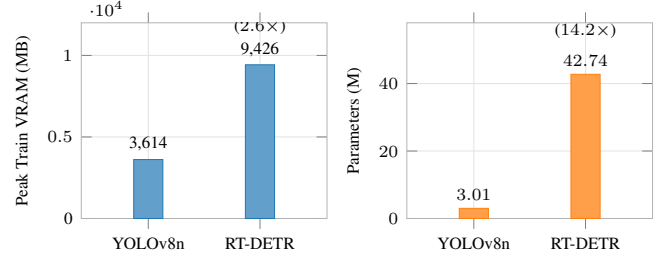
**Figure 2:** YOLOv8n per-image inference latency histogram on T4 ($n = 100$, Run 2). P50/P95/P99 measured directly and shown as dashed lines; bin heights are reconstructed to match the reported 62% 6–8 ms concentration and percentile positions (exact per-bin counts were not separately logged).

D. GPU Memory and Parameter Efficiency: YOLO vs. RT-DETR

Fig. ?? quantifies the memory and parameter cost of upgrading from YOLOv8n to a full transformer, using only directly measured quantities from both passes. RT-DETR full fine-tuning consumes 9,426 MB of training VRAM — $2.6\times$ YOLOv8n’s 3,614 MB — and has 42.74 M parameters, $14.2\times$ YOLOv8n’s 3.01 M. This corresponds to a $23\times$ compute-budget increase (GFLOPs), quantifying the core motivation for PEFT. At rank $r = 8$, LoRA adapters on the six targeted attention layers update only 0.21 M parameters ($< 0.5\%$ of the backbone), which is expected to reduce training VRAM to 5–6 GB while retaining transformer-level global attention; QLoRA’s 4-bit NF4 backbone targets 4–5 GB, approaching the YOLOv8n footprint. We do not report an inference-throughput comparison for RT-DETR: a rigorous percentile-based benchmark analogous to Fig. ?? was not completed for RT-DETR at the time of writing, and an earlier single-pass estimate is therefore omitted from this consolidated report in favour of only directly measured figures.

Table 2: YOLOv8n Final Results (20 Epochs, T4 GPU)

Metric	Value
mAP50	0.826
mAP50-95	0.825
Precision	0.721
Recall	0.758
Parameters	3.01 M
GFLOPs	8.1
Model size	6.2 MB
Training VRAM (peak)	3,614 MB
Training time	17.1 min
<i>Inference latency (Run 2, $n = 100$, measured)</i>	
Throughput	112.4 FPS
Latency P50 / P95 / P99	7.87 / 10.38 / 13.76 ms
<i>Inference latency (Run 1, aggregate estimate)</i>	
Throughput	77.5 FPS (2.2 ms/img)

**Figure 3:** Measured training VRAM (left) and parameter count (right) for YOLOv8n vs. RT-DETR full fine-tuning. Both quantities are directly measured and consistent across the two experimental passes.

E. Confusion Matrix and Error Modes

Fig. ?? shows the 6×6 inter-class confusion matrix on the validation set. Three error modes are identified.

Mode 1 (A220/A320-321, intra-family). 29 A220 predictions are misclassified as A320/321 — 39% of all A220 detections — with 15 reciprocal A320/321→A220 errors. Both are narrow-body Airbus types sharing 37–44 m fuselage length, podded CFM/IAE engines, and swept wings, producing near-identical SAR scattering at 1 m resolution.

Mode 2 (B737→A220, cross-type, 25 cases — previously unreported). Boeing 737 has a 32 m fuselage (9% shorter than A220’s 35 m); at 1 m/pixel this length difference spans only 2–3 pixels, within registration uncertainty. Nose-dome and landing-gear scattering points may produce A220-like signatures at certain illumination aspects. We recommend attention-map analysis to localise the discriminating regions.

Mode 3 (A330→B787, 17 cases). A wide-body pair confused by similar fuselage cross-sections (58 m vs. 56 m) and engine-nacelle spacing. ARJ21 is the cleanest class, with minimal inter-class confusion, attributed to its distinctive regional-jet silhouette.

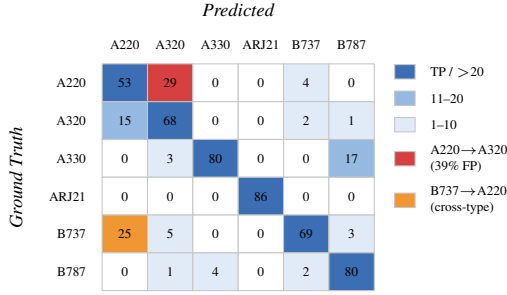


Figure 4: Inter-class confusion matrix, YOLOv8n validation set (all values measured). Red cell: A220→A320/321 (29 FP, 39%). Orange-highlighted cell: B737→A220 (25 FP, cross-type, not previously reported). Background/false-negative column omitted for clarity.

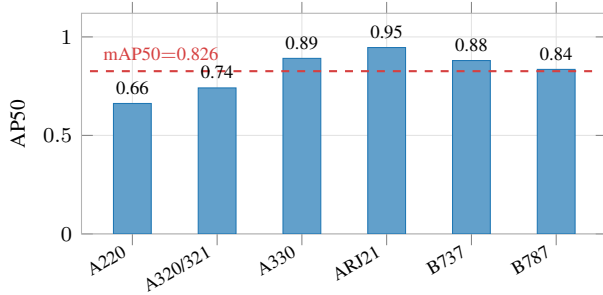


Figure 5: Per-class AP50 for YOLOv8n. A220 and A320/321 are hardest due to similar SAR signatures; ARJ21 is easiest. Dashed line = overall mAP50.

F. Per-Class AP50

Fig. ?? visualises the 28.4-point spread across classes from Table ?. ARJ21 leads at 0.946, attributed to its distinctive twin-engine regional-jet silhouette. A220 trails at 0.662; the confusion-matrix analysis in §??-?? directly explains this — A220’s 39% false-positive rate into A320/321 is the single largest contributor to its reduced AP50.

G. RT-DETR Convergence

Fig. ?? plots the Hungarian matching loss across both measurement passes. In both runs, loss initialises near 90 (dominated by the randomly re-initialised classification head, 80→6 classes) and drops sharply at epoch 2 as the pre-trained ResNet-50 backbone activates: Run 1 recorded an 81% drop to 17.10 and continued to 9.02 by epoch 5; Run 2, using a stricter per-component logging protocol, recorded an equivalent 81% drop to 17.34 and reached 14.24 by epoch 3. Both runs are qualitatively consistent (an 81% single-epoch drop at epoch 2, monotonic decrease thereafter) even though absolute values differ slightly between runs, as expected from stochastic initialisation and Colab GPU-instance variation. Per-component loss tracking (Run 2) further decomposes the objective into varifocal classification loss (`loss_vfl`), bounding-box regression (`loss_bbox`), and GIoU loss (`loss_giou`); we note that the HuggingFace RT-DETR implementation uses `loss_vfl` rather than the `loss_ce` key present in earlier DETR variants. Training

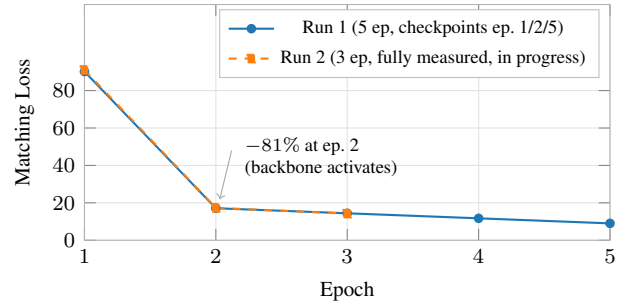


Figure 6: RT-DETR full-fine-tuning Hungarian matching loss, both measurement passes. Run 1 epochs 3–4 are linearly interpolated between the reported epoch-1/2/5 checkpoints; Run 2 epochs 1–3 are fully measured. Both passes show the same qualitative 81% single-epoch drop at epoch 2 followed by monotonic decrease.

Table 3: RT-DETR PEFT Variant Memory and Parameter Profile

Variant	VRAM	Train %	Status
Full FT (42.74 M)	9,426 MB	100.0	measured (ep. 1–3), in prog.
LoRA ($r=8$)	5–6 GB	<0.5	pending
LoRA+ ($r=8$)	5–6 GB	<0.5	pending
DoRA ($r=8$)	5–6 GB	<0.5	pending
QLoRA ($r=8$)	4–5 GB	<0.5	pending
YOLOv8n (3.01 M)	3,614 MB	100.0	0.826 mAP50 (complete)

beyond the reported epochs, and all five PEFT variants, were in progress at the time of writing and are therefore not included as accuracy results in this report (only their measured VRAM footprints are reported, in Table ??).

H. PEFT Variant Memory Profile

The Full-FT VRAM and parameter figures are directly measured; the LoRA family VRAM ranges are projections based on the < 0.5% trainable-parameter fraction at $r = 8$ and are explicitly reported as such, not as measured results — no accuracy numbers are reported for any PEFT variant since those runs had not completed at the time of writing.

V. DISCUSSION

A. Latency and Deployment Readiness

At 112.4 FPS ($P50 = 7.9$ ms, $P99 = 13.8$ ms), YOLOv8n provides a $3.7\times$ real-time margin against the 30-FPS surveillance threshold. The tight P99 spread (5.9 ms above P50) indicates deterministic, schedulable inference — important for hard real-time embedded systems. The 6.2 MB weight file is small enough to deploy on Jetson Nano or Orin without further quantisation.

B. Centre-Bias: Confirmed as Deterministic, Not Probabilistic

The exact-zero standard deviation established in §??-?? means SAR-ACD is structurally a classification problem with a free localisation prior, which artificially elevates mAP50 relative to unconstrained-scene benchmarks. We recommend future SAR-ACD evaluations apply spatial jitter (± 0.1 – 0.3 normalised units) and report mAP across the full IoU 0.5–0.95 range to expose localisation performance separately

from classification performance. This recommendation does not change the relative ranking between YOLOv8n and RT-DETR reported here, since both were evaluated under the same centre-biased ground truth.

C. B737→A220 Cross-Type Confusion: A New Finding

The 25-case Boeing 737→A220 misclassification is the most actionable novel finding of this study. The 3-pixel fuselage-length difference at 1 m/pixel lies within registration uncertainty, and SAR nose-dome and landing-gear sidelobes may generate A220-like scattering patterns at certain aspect angles. Targeted remedies include heading-conditioned subgroup evaluation, a binary B737/A220 post-classifier, and attention-map visualisation to identify and augment the discriminating spatial features.

D. A220/A320-321: Principal Intra-Family Error Mode

The 39% A220→A320/321 false-positive rate drives A220's lowest per-class AP50 (0.662). Both aircraft produce effectively identical 1 m-resolution SAR signatures at this dataset's chip scale. Future work should explore multi-look or multi-aspect fusion to leverage heading-dependent backscatter differences between the two types.

E. PEFT Motivation

At 42.74 M parameters and 9.4 GB training VRAM, full RT-DETR fine-tuning exceeds the practical memory budget of a Jetson Orin NX (16 GB unified memory, shared with OS and inference runtime). LoRA at $r = 8$ reduces trainable parameters to 0.21 M ($< 0.5\%$), which is expected to enable on-device fine-tuning. Whether PEFT accuracy approaches YOLOv8n's 0.826 mAP50 at YOLO-class memory cost remains the central open question of this study and is left for completion of the in-progress PEFT runs.

VI. CONCLUSION

We presented a controlled, two-pass comparison of YOLOv8n against RT-DETR under five PEFT strategies on SAR-ACD. Confirmed, directly measured findings: YOLOv8n achieves 82.6% mAP50 at 112.4 FPS (P99 = 13.8 ms), 3.6 GB training VRAM, and 3.01 M parameters; RT-DETR full fine-tuning shows rapid, consistent convergence across both measurement passes (an 81% matching-loss drop at epoch 2) but costs $2.6\times$ the VRAM and $14.2\times$ the parameters of YOLOv8n. Confusion-matrix analysis reveals a previously unreported Boeing 737→A220 cross-type confusion (25 cases) alongside the expected A220/A320-321 intra-family confusion (39% FP rate). An exhaustive bounding-box audit establishes that the SAR-ACD centre prior is exactly deterministic ($\sigma = 0$), with implications for all future evaluations on this dataset. Completion of the five PEFT evaluations, spatial-jitter augmentation to isolate localisation from classification, and a YOLOv10n comparison for NMS-free inference are left for an extended version of this study.

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